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Diazaphospholenes as reducing agents: a thermodynamic and electrochemical DFT study†

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Diazaphospholenes have emerged as a promising class of metal-free hydride donors and have been implemented as molecular catalysts in several reduction reactions. Recent studies have also verified their radical reactivity as hydrogen atom donors. Experimental quantification of the hydricities and electrochemical properties of this unique class of hydrides has been limited by their sensitivity towards oxidation in open air and moist environments. Here, we implement quantum chemical density functional theory calculations to analyze the electrochemical catalytic cycle of diazaphospholenes in acetonitrile. We report computed hydricities, reduction potentials, pK_a values, and bond dissociation free energies (BDFEs) for 64 P-based hydridic catalysts generated by functionalizing 8 main structures with 8 different electron donating/withdrawing groups. Our results demonstrate that a wide range of hydricities (29–66 kcal mol^{-1}) and BDFEs (58–81 kcal mol^{-1}) are attainable by functionalizing diazaphospholenes. Compared to the more common carbon-based hydrides, diazaphospholenes are predicted to require less negative reduction potentials to electrochemically regenerate hydrides with an equivalent hydridic strength, indicating their higher energy efficiency in the tradeoff between thermodynamic ability and reduction potential. We show that the tradeoff between the reducing ability and the energetic cost of regeneration can be optimized by varying the BDFE and the reorganization energy associated with hydride transfer (λ_{HT}), where lower BDFE and λ_{HT} correspond to more efficient catalysts. Aromatic phosphorus hydrides with predicted BDFEs of ~62 kcal mol^{-1} and λ_{HT}'s of ~20 kcal mol^{-1} are found to require less negative reduction potentials than dihydropyridines and benzimidazoles with predicted BDFEs of ~68 and ~84 kcal mol^{-1} and λ_{HT}'s of ~40 and ~50 kcal mol^{-1}, respectively.

Introduction

Hydride transfer (HT) can efficiently reduce electrophilic species by avoiding high energy open-shell intermediates produced by single electron reductions because they are equivalent to transferring two electrons and a proton. Consequently, HT reactions represent a promising class of redox reactions. In HTs, the hydride donor is oxidized by donating a hydride anion to a desired acceptor. Some classes of hydride donors can be strong reducing agents, which makes them attractive for catalytic applications because this affords them the ability to activate very stable molecules such as CO_2. In addition, hydride donors play a major role in synthetic chemistry.[1–3] Transition metal hydrides are, in general, excellent hydride donors that can be used in catalytic quantities.[4–7] However, due to their lower abundancy and high toxicity,[8] recent efforts have shifted towards exploring organic-based hydride donors as alternatives to metal-based hydrides.[4]

Several studies have shown that some nonmetal-based hydride donors are sufficiently strong that they reduce molecules containing stable bonds such as C═O, C═N and C═C.[1,2] Nonmetal-based hydrides contain one hydridic H atom attached either to a nonmetal (such as carbon, nitrogen, oxygen, phosphorus and sulfur) or a metalloid (such as boron, silicon, germanium and tellurium).[4] The strength of a hydride can be quantified by its thermodynamic and kinetic ability. The thermodynamic hydricity (ΔG_{H^-}) is simply the change in Gibbs free energy of the reaction associated with donating the hydride:

$$XH \rightarrow X^+ + H^- \quad (1)$$

where lower ΔG_{H^-} values correspond to stronger hydride donors. In acetonitrile, a hydride donor is considered to be of moderate strength if its ΔG_{H^-} value lies between 50–80 kcal mol^{-1}.[4]

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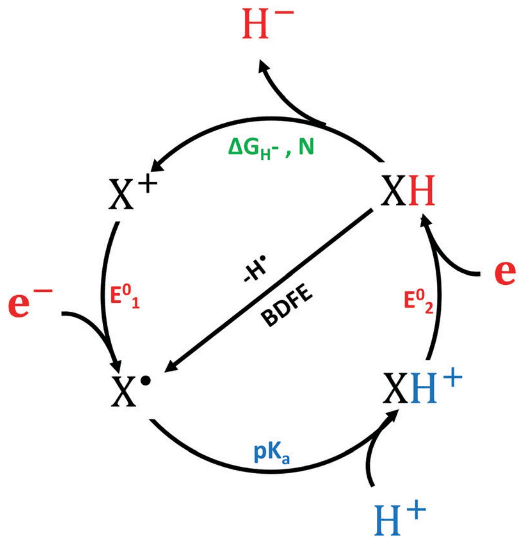
The hydricities of nonmetal-based hydride donors can be modulated over a wide range by manipulating the electronic and structural factors that influence the free energy of the HT reaction. For organic hydrides that become aromatic upon HT, the main driving force for hydride donation is aromatization.[9] Carbon-based hydrides constitute most of the common organic hydrides. In biological systems, for example, nicotinamide adenine dinucleotide phosphate (NADPH) plays a central role in CO_2 reduction through a HT mechanism.[10] Some other common classes of carbon-based hydride donors include pyridines, pteridines, and benzimidazoles.[11–21] The majority of nonmetal-based hydride donors are moderate or weak hydrides with few exceptions, such as some radical anion and boron-based hydrides.[4]

Phosphorus-based hydrides (PBHs) have recently emerged as strong organic hydride donors.[22–24] They have been implemented as molecular catalysts in several applications such as: hydrogenation of N=N bonds with ammonia–borane,[25] hydroboration of pyridines[26] and CO_2 reduction to formate.[27] Despite the well-known fact that the bonds between hydrogen and group 15 elements are usually protic in nature, including bonds between hydrogen and phosphorus, the heterocyclic structure of diazaphospholene results in its unique hydridic P–H bond.[24] Furthermore, the fact that phosphorus and hydrogen have equal electronegativities has led to the tendency of this class of molecules to act as reducing agents not only by hydride donation but also through a hydrogen atom transfer mechanism.[28] The strength of these hydride donors and their increasing applicability, while being organic and thus relatively inexpensive, has increased the motivation to evaluate their catalytic performance to provide a framework for identifying optimal PBHs to catalyze efficient industrial processes. A recent study experimentally investigated the reaction kinetics of HT from PBHs to some common electrophiles and established a nucleophilicity scale.[29] Although this can be useful in comparing the relative strengths of different PBHs, there remains a dearth of information related to their thermodynamic and electrochemical properties, which are crucial for identifying specific applications for these hydride donors.

In addition to analyzing the thermodynamics and kinetic barriers of HT, the process of catalyst regeneration and its efficiency must be considered to evaluate the catalyst's performance. Scheme 1 shows the electrochemical catalytic cycle of molecular hydrides and the associated metrics for each step: thermodynamic hydricities for HTs, reduction potentials for electron transfers, pK_a values for proton transfers, and hydrogen atom bond dissociation free energies (BDFEs) for H atom transfers. A clear understanding of all these metrics is required to form a complete picture of the feasibility and applicability of implementing a candidate molecular hydride in a catalytic process. In many cases, it is not convenient to determine these parameters experimentally. For example, attempts to experimentally determine the thermodynamic hydricities of N-heterocyclic phosphines, which are also sensitive towards air and moisture, have been hindered by the lack of proper counter ions that are necessary to achieve equilibrium.[30] In such cases, quantum chemical calculations are an attractive



Scheme 1 Electrochemical catalytic cycle of hydride/hydrogen atom donors. [XH]: the hydride donor. [X^+]: the cation resulting from hydride donation. [$X^\bullet$]: the radical intermediate produced by the first reduction step. [XH^+]: the radical cation resulting from the protonation step.[14]

alternative, especially for screening a wide range and large number of candidate molecules. One technique to avoid this issue experimentally, is to start by using alkoxydiazaphospholene as a catalyst precursor that can be handled in open air and then regenerating the P–H bond from the P–O bond.[31]

In this work we use quantum chemistry based on density functional theory (DFT) to evaluate the catalytic performance of PBHs based on both their hydridic strength and regeneration efficiency. We explored 8 general PBH structures (Fig. 1). Each structure is substituted with 4 electron-donating groups (–NH_2, –OH, –CH_3 and –*t*Butyl) and 3 electron-withdrawing groups (–CN, –COOH and –Cl). This forms a set of 64 possible P-based hydridic catalysts, including –H terminations. We employ quantum chemical computations to overcome the experimental limitations of evaluating the electrochemical catalytic cycle. Our aim is to use this evaluation to provide a better understanding of the hydridic and the electrochemical characteristics of this class of molecules to enable a shift in the capabilities of

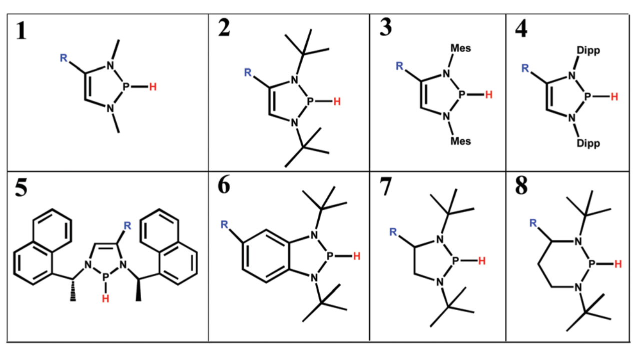


Fig. 1 Structures of P-based hydrides. Mes = 2,4,6-$Me_3C_6H_2$ and Dipp = 2,6-$iPr_2C_6H_3$. Note that **7** and **8** have saturated rings and are thus non-aromatic upon loss of the hydride.

organic hydrides from moderately strong hydrides to strong hydrides capable of inducing challenging reductions while being reasonably efficient to regenerate. Furthermore, these results can guide the future implementation of phosphorus hydrides in different chemistries by providing detailed scales of their thermodynamic and electrochemical properties.

Computational methodology

The M06 exchange correlation functional combined with the 6-31+G(d,p) basis set is utilized to optimize structures and calculate ground state free energies in acetonitrile. This level of theory describes the system accurately at a moderate computational cost because the M06 functional has been parameterized with experimental thermodynamic data for similar molecular systems.[32] We used the SMD universal solvation model in this study to account for the solvation effects. SMD is based on the solute electron density and a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions.[33] Benchmarking of DFT functionals, basis sets and solvation models is described in the ESI.† While the level of theory does not significantly change the computed properties, the solvation model does, which justified our choice of the SMD model, which is also recommended by the Gaussian 16© software package we used to perform the calculations.[34]

Hydricity

The thermodynamic hydricity is calculated by taking the difference in the free energies of reactants and products of the HT reaction (eqn (1)):

$$\Delta G_{\mathrm{H}^-} = G_{\mathrm{H}^-} + G_{\mathrm{X}^+} - G_{\mathrm{XH}} \tag{2}$$

While both G_{XH} and G_{X^+} can be determined directly using DFT, the free energy of the solvated hydride (G_{H^-}) cannot be calculated accurately using a combination of existing DFT and implicit solvation models.[35–38] Alternatively, G_{H^-} can be estimated by using the isodesmic approach,[39,40] which involves calibration against a reference, or by using the linear scaling approach,[41] which involves calibration against multiple reference hydrides. In this work, we adopt the isodesmic approach due to the dearth of available experimental hydricities of diazaphospholenes. Because the hydricity of molecule **8-H** has been determined experimentally to be 48.8 kcal mol^{-1},[28] we selected it as a reference hydride and accordingly, estimated G_{H^-} to be -405.1 kcal mol^{-1}, in agreement with previous results.[4]

Reduction potential

The reduction potential of the X^+ intermediate is a key factor that determines the thermodynamic efficiency, and thus economic viability, of catalyst regeneration. Here, we report reduction potentials *vs.* the ferricenium/ferrocene (Fc^+/Fc) reference electrode, which is the IUPAC recommended reference electrode for non-aqueous solutions.[42] Reduction potentials are calculated using the following equation:

$$E^0 = G_{\mathrm{X}^+} - G_{\mathrm{X}^\bullet} - E^0_{\mathrm{ref}} \tag{3}$$

where E^0_{ref} is the absolute redox potential of the reference electrode (4.988 V for Fc^+/Fc in acetonitrile)[43] and both G_{X^+} and $G_{\mathrm{X}^\bullet}$ are in eV. We calculate $E^0_1 = -2.44$ V for **8-H**, which is in good agreement with the experimental value of -2.39 V.[28]

pK_a

Protonation follows ET and is thus the second step in the regeneration process. The tendency of an intermediate $\mathrm{X}^\bullet$ to accept a proton is quantified by the pK_a value of its protonated form XH^+. Higher pK_a's indicate more thermodynamically favorable protonation of the $\mathrm{X}^\bullet$ intermediate and thus more efficient catalyst regeneration. pK_a values are calculated as follows:

$$\mathrm{p}K_\mathrm{a} = \frac{\Delta G_{\mathrm{H}^+}}{2.303RT} \tag{4}$$

$$\Delta G_{\mathrm{H}^+} = G_{\mathrm{H}^+} + G_{\mathrm{X}^\bullet} - G_{\mathrm{XH}^+} \tag{5}$$

while $G_{\mathrm{X}^\bullet}$ and G_{XH^+} are calculated using DFT, and G_{H^+} is determined experimentally to be -254.2 kcal mol^{-1}.[44]

BDFE

The bond dissociation free energy is the free energy change associated with the homolytic dissociation of the XH bond:

$$\mathrm{XH} \rightarrow \mathrm{X}^\bullet + \mathrm{H}^\bullet \tag{6}$$

The tendency of XH to donate a hydrogen atom is a maximum for low BDFEs. We calculate BDFEs using the following equation:[45,46]

$$\begin{aligned}\mathrm{BDFE} &= G_{\mathrm{H}^\bullet} + G_{\mathrm{X}^\bullet} - G_{\mathrm{XH}} = \Delta G_{\mathrm{H}^-} - F\left(E^0_1 - E^{\mathrm{H}^-}_{\mathrm{ox}}\right) \\ &= \Delta G_{\mathrm{H}^-} - 23.06E^0_1 - 26.0\end{aligned} \tag{7}$$

where F is Faraday's constant and $E^{\mathrm{H}^-}_{\mathrm{ox}}$ is the oxidation potential of H^-.

Our calculated BDFE for **8-H** (79.0 kcal mol^{-1}) agrees with the experimentally measured BDFE of 77.9 $\pm$ 2.0 kcal mol^{-1}.[28]

Reorganization energy

In this work, we implement the HT reorganization energy (λ_{HT}) as a unique metric to evaluate the catalytic efficiency of organic hydrides. λ_{HT} is calculated as follows:

$$\lambda_{\mathrm{HT}} = G_{\mathrm{X}^+} - G'_{\mathrm{X}^+} \tag{8}$$

where G'_{X^+} is the free energy of the cation X^+ computed using a single-point energy calculation at the fixed geometry of the original structure XH excluding the hydride.

Results and discussion

The results listed in Table 1 predict that PBHs span a wide range of hydricities (29–66 kcal mol^{-1}) that can be tuned by modifying the heterocyclic structure of the backbone or by functionalizing it with appropriate functional groups. While PBH molecular structures **1–6** become aromatic as they donate their hydrides, **7** and **8** and their derivatives consist of saturated, non-aromatic ring structures. The most common PBHs, which

Table 1 Computed hydricities ΔG_{H^-} (kcal mol^{-1}), reduction potentials E^0 (V *vs.* Fc$^+$/Fc), pK_a values, and bond dissociation free energies (kcal mol^{-1}) in acetonitrile

XH	R	ΔG_{H^-}	E_1^0	E_2^0	pK_a	BDFE	XH	R	ΔG_{H^-}	E_1^0	E_2^0	pK_a	BDFE
1	H	36.4	−2.23	−1.38	17.2	61.8	**5**	H	38.1	−2.15	−1.25	15.1	61.6
	CH_3	34.8	−2.27	−1.51	19.2	61.2		CH_3	36.6	−2.20	−1.44	17.6	61.3
	tBut	36.0	−2.28	−1.34	18.6	62.5		tBut	40.6	−2.22	−1.48	17.0	65.8
	NH_2	31.9	−2.36	−1.85	23.6	60.3		NH_2	34.2	−2.31	−1.81	22.5	61.6
	OH	36.2	−2.23	−1.57	19.4	61.7		OH	37.5	−2.19	−1.50	18.2	62.1
	CN	47.8	−1.78	−0.77	9.4	62.8		CN	52.7	−1.73	−0.73	8.2	66.7
	COOH	46.9	−1.91	−0.93	11.2	64.9		COOH	49.5	−1.80	−0.93	10.3	64.9
	Cl	40.9	−2.08	−1.28	15.0	63.0		Cl	44.2	−1.97	−1.26	13.5	63.8
2	H	34.1	−2.35	−1.46	17.4	62.3	**6**	H	44.5	−1.96	−0.78	12.3	63.7
	CH_3	32.4	−2.41	−1.54	19.8	61.9		CH_3	42.7	−2.05	−0.90	14.5	64.1
	tBut	33.8	−2.43	−1.50	19.0	63.8		tBut	42.6	−2.00	−0.89	13.4	62.8
	NH_2	29.1	−2.52	−1.89	24.4	61.2		NH_2	40.7	−2.07	−1.15	18.2	62.3
	OH	31.9	−2.49	−1.65	20.6	63.3		OH	43.0	−1.97	−0.97	14.6	62.6
	CN	45.6	−1.85	−0.87	10.8	62.3		CN	49.3	−1.69	−0.46	7.7	62.3
	COOH	44.9	−2.07	−1.04	12.0	66.6		COOH	52.3	−1.77	−0.60	12.0	67.2
	Cl	37.8	−2.27	−1.28	15.7	64.1		Cl	45.9	−1.84	−0.67	10.5	62.4
3	H	41.4	−2.02	−1.13	14.8	61.9	**7**	H	47.3	−2.14	−0.60	8.1	70.7
	CH_3	39.5	−2.02	−1.28	15.7	60.1		CH_3	46.5	−2.14	−0.63	6.9	69.9
	tBut	39.7	−2.19	−1.22	15.9	64.2		tBut	51.8	−2.15	−0.50	7.9	75.4
	NH_2	35.5	−2.19	−1.62	21.6	60.1		NH_2	46.9	−2.11	−0.62	6.3	69.5
	OH	39.1	−2.21	−1.41	16.7	64.1		OH	49.3	−2.04	−0.48	7.3	70.3
	CN	53.5	−1.52	−0.38	5.4	62.6		CN	54.2	−1.80	−0.29	4.2	69.6
	COOH	57.1	−1.57	−0.65	12.2	67.3		COOH	57.2	−2.00	−0.22	6.9	77.2
	Cl	46.7	−1.90	−1.00	12.3	64.5		Cl	66.0	−1.73	−0.13	−4.2	79.8
4	H	42.4	−1.92	−0.99	12.6	60.7	**8**	H	48.8	−2.44	−0.91	10.6	79.0
	CH_3	41.2	−2.07	−1.12	13.9	62.9		CH_3	49.5	−2.38	−0.90	11.7	78.3
	tBut	44.3	−2.06	−1.17	14.9	65.7		tBut	51.6	−2.31	−0.87	10.9	78.8
	NH_2	38.1	−2.02	−1.48	19.6	58.6		NH_2	50.7	−2.34	−0.88	11.2	78.7
	OH	41.9	−1.95	−1.20	15.2	61.0		OH	51.4	−2.32	−0.81	11.4	78.9
	CN	56.2	−1.49	−0.35	4.3	64.5		CN	55.9	−2.19	−0.72	7.3	80.4
	COOH	47.8	−1.59	−0.57	4.8	58.5		COOH	51.8	−2.24	−0.72	9.7	77.4
	Cl	48.7	−1.79	−0.82	10.0	63.9		Cl	58.5	−2.06	−0.60	−2.3	80.0

are investigated in this study, are, in general, strong hydride donors, which can be confidently concluded from their computed hydricity values reported in Table 1. In fact, molecules such as **2** have been considered to be unusually strong hydrides, which explains why these hydrides have been successfully used to reduce highly stable molecules such as CO_2.[27] Moreover, aromatic hydrides (molecules **1–6**) are generally stronger hydride donors than their non-aromatic counterparts. Indeed, aromatization is a significant driving force for hydride donation of many organic hydrides.[9,13,16] Aromatic hydrides become weaker as the size of their π systems increases. This results from competing effects of delocalization caused by the addition of fused ring(s) where an inductive effect weakens the hydride by withdrawing electron density more than resonance stabilizes the product of hydride transfer.[14]

While stronger hydrides enable more challenging reductions, this comes at the expense of their regeneration cost. Fig. 2 demonstrates the tradeoff between hydricity and the first reduction potential, which is in turn the major contributor towards the regeneration cost. This kind of relationship was reported for both metal-free and metal-based hydrides.[4,14] While lower hydricity values are preferred for certain applications, less negative reduction potentials are needed in order to make the catalytic process feasible. Thus, in this study we aimed to provide

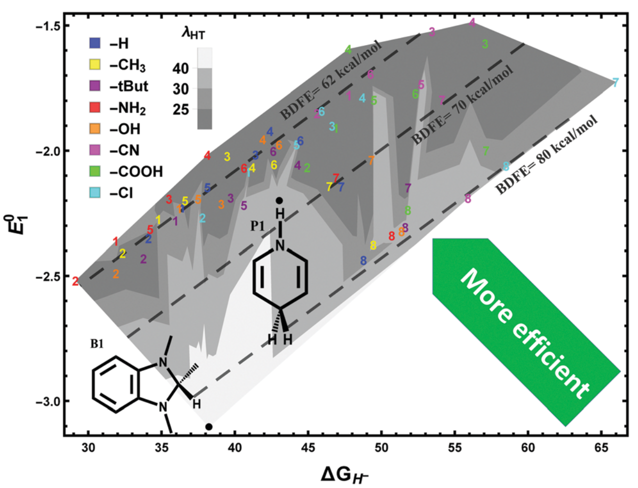


Fig. 2 First reduction potential E_1^0 [V] (*vs.* Fc$^+$/Fc) *vs.* hydricity ΔG_{H^-} [kcal mol^{-1}] of P-based hydrides, a benzimidazole hydride (**B1**) and dihydropyridine (**P1**) in acetonitrile. The plot illustrates the shift in the tradeoff relationship as a result of shifting BDFE which is also associated with variation in the reorganization energy (λ_{HT}) associated with HT. Lower hydricity indicates stronger hydride and less negative reduction potential corresponds to higher regeneration efficiency.

wide ranges of hydricities and reduction potentials by starting with hydrides of different strengths and functionalizing them with both electron-withdrawing groups (EWGs) and electron-donating groups (EDGs). The second reduction potential is of less importance for the regeneration efficiency as it is always significantly less negative than the first reduction potential. This arises because the products of the first reduction step are high energy open-shell, radical species, whereas the second reduction step results in more stable, low energy closed-shell species. The relationship between hydricity and reduction potential results in multiple groupings of ΔG_{H^-} *vs.* E_1^0 [V] in the plot illustrated in Fig. 2: aromatic hydrides (**1–6**), and non-aromatic hydrides **7** and **8**. Two examples of C-based hydrides, **P1** (dihydropyridine) and **B1** (a benzimidazole), are shown for comparison.

Our calculations indicate that at similar hydricity values, the first reduction step of PBHs occurs at a significantly less negative potential relative to the first reduction potential of C-based hydrides. The shift in this limiting relationship corresponds to differences in the BDFE across different classes of hydride donors. Because BDFE is equivalent to the sum of the energy required to donate the hydride and the energy needed to transfer one electron, as shown in Scheme 1, reducing its magnitude results in both lower hydricities and less negative reduction potentials, thus making the catalytic hydride donation more efficient. Notably, the shifts in BDFE between different classes of molecular hydrides are accompanied by significant variations in the reorganization energy associated with HT. Larger λ_{HT} indicates that the hydride donor undergoes a significant change in its geometry as it donates the hydride. This results in a more stable cationic species (X^+) that requires a more negative reduction potential to regenerate its corresponding hydride, thus making the hydride a less efficient catalyst. The stability of the cationic counterparts is a major contributor to the hydridic strength of C-based and non-aromatic P-based hydrides, while aromatic PBHs remain unstable even after donating the hydride, which explains the difficulty in measuring their hydricities experimentally by establishing equilibrium with a reference hydride acceptor.[30] Similarly, large BDFEs correspond to unstable radical intermediates ($X^•$) that result in more negative reduction potentials. The coupling between cationic and radical intermediate stabilization most likely arises from the resonance effect, which tends to stabilize both the positive charge and the transferred electron. Aromatic PBHs exhibit less negative reduction potentials as a result of their low BDFEs (~62 kcal mol^{-1}), which correspond to their enhanced resonance stabilization, and their low reorganization energies (~20 kcal mol^{-1}) that correspond to their rigid planar structures. Derivatives of **7** are more flexible than the aromatic hydrides due to the absence of double bonds on the ring structure, which leads to a higher reorganization energy. They also have fewer resonance structures compared to aromatic PBHs. On the other hand, derivatives of **8** have even larger BDFEs (~79 kcal mol^{-1}) and reorganization energies (~30 kcal mol^{-1}) as their larger ring size enables more geometric distortions compared to derivatives of **7**. Our results predict that aromatic phosphorus hydrides outperform dihydropyridines (BDFE ~ 68 kcal mol^{-1} and λ_{HT} ~ 40 kcal mol^{-1}), while benzimidazoles (BDFE ~ 84 kcal mol^{-1} and λ_{HT} ~ 50 kcal mol^{-1}) have the worst performance in the tradeoff between reduction potential and hydricity. As a result of their significantly lower λ_{HT} values, we expect PBHs to be also more kinetically active relative to their C-based counterparts at the same thermodynamic driving force as they are closer to an ideal hydride donor (λ_{HT} = 0) – one with no reorganization energy penalty due to hydride transfer.[47] We computed an activation energy of 17.2 kcal mol^{-1} for the HT reaction of **5-H** with CO_2, whereas the C-based benzimidazole hydride **B1** has an activation energy of 25.5 kcal mol^{-1}.

Functional group effects

Functionalizing organic hydrides with different functional groups redistributes the electron density in the conjugated π system through resonance and/or induction. While EDGs increase the electron density, EWGs have the opposite effect,

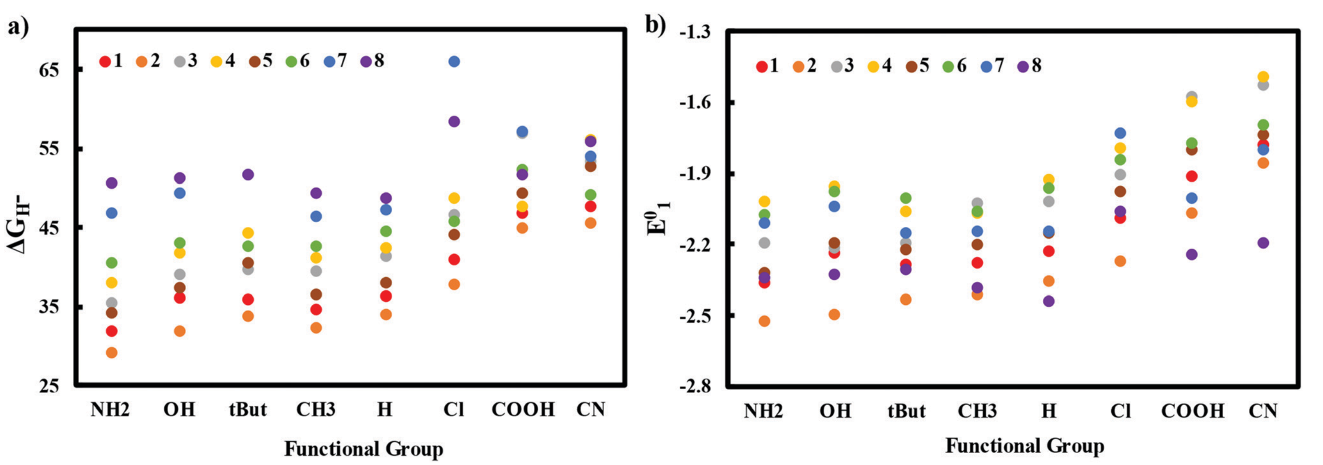


Fig. 3 Effect of EDGs and EWGs on: (a) hydricity ΔG_{H^-}, (b) 1st reduction potential E_1^0 (*vs.* Fc^+/Fc). Functional groups implemented in this study are listed in the following order from most donating to most withdrawing (based on their Hammett constants): $-NH_2$, –OH, –*t*Butyl, $-CH_3$, –H, –Cl, –COOH and –CN.[48] The Hammett Constant is a substituent-specific constant that quantifies the extent of electron donation/withdrawl.[49] More negative Hammett constants correspond to maximum electron donation and *vice versa*. Here we sort functional groups based on the Hammett constants of substituted benzene rings.

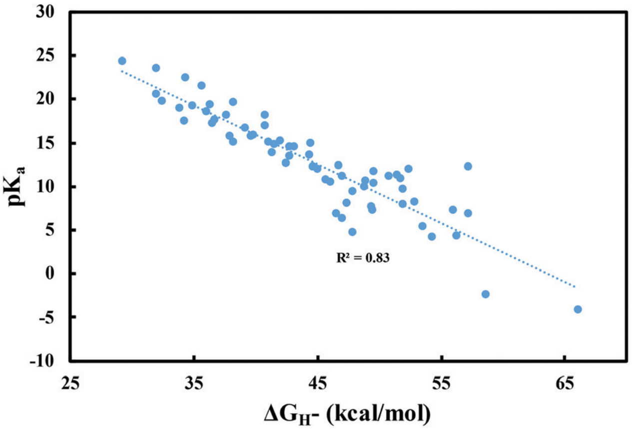


Fig. 4 pK_a value of XH^+ *vs.* hydricity for P-based hydrides in acetonitrile. Higher pK_a values correspond to easier protonation. The plot indicates that more acidic XH^+ intermediates correlate with weaker subsequent hydride donors.

thus enabling the modulation of the hydride strength by functionalization. Increasing the electron density of the hydride enhances its hydridic strength, which makes it possible to strengthen weak and moderate hydrides by substituting them with EDGs. However, this introduces a disadvantage when implementing these hydrides as HT catalysts because the reduction step in their regeneration will now require a more negative reduction potential and thus be less efficient. On the other hand, EWGs affect hydride donors in the opposite way. While they weaken the HT driving force of the hydride by lowering its electron density, they make the implementation of the hydride as a catalyst more efficient as they require less negative reduction potentials for their regeneration.[14]

Fig. 3 shows the effect of electron donation/withdrawal on the hydricity and first reduction potential. The general trend is that lower electron donation correlates with larger hydricities (weaker hydrides) and less negative reduction potentials. There are a few exceptions to this general trend. For example, although EDGs tend to increase the driving force for HT, it is apparent that they can paradoxically slightly reduce the hydridic strength in some cases. This counterintuitive effect mainly results from steric hindrance that creates an energetic penalty to HT as transfer of the hydride anion (H^-) to the acceptor becomes hindered in bulky molecules such as **4** and **5**. On the other hand, the trends of hydricity and E^0 with functionalization are also less systematic for non-aromatic hydrides because their π system of electrons is only partially conjugated, which limits the effect of the functional group.

Protonation of X•

Fig. 4 shows that the pK_a value of XH^+ is approximately linearly proportional to the hydricity of XH. Most of the hydrides studied in this work have moderate to high pK_a values, which indicates that their protonation step can be facilely accomplished with a mild proton source. Furthermore, the linear relation reveals that as the hydride becomes stronger, the protonation step becomes more facile. Unlike the correlation between hydricity and reduction potential, the relationship between hydricity and pK_a values does not depend on the class of PBH (aromatic or non-aromatic) as the protonation step does not involve aromatization or de-aromatization and the reorganization energy associated with it does not vary significantly between different candidates.

Hydrogen atom donation

In addition to HT reactions, diazaphospholenes can also donate a hydrogen atom to H atom acceptors with X-H BDFEs of more than 60 kcal mol^{-1}. This radical reactivity has enabled the implementation of diazaphospholenes in hydrodebromination and hydrodechlorination reactions of aryl and alkyl halides, and chemoselective cleavage of the α-C–O bonds in α-carboxy ketones.[30,50,51] Fig. 5a shows the effect of EDGs and EWGs on the tendency of PBHs to donate hydrogen atoms. These results indicate that the electron density in the ring does not significantly impact the BDFE. This corroborates the reported results that rule out any effect of increasing the size of the π system of electrons, which is equivalent to adding an EWG, on BDFE.[28] This is due to

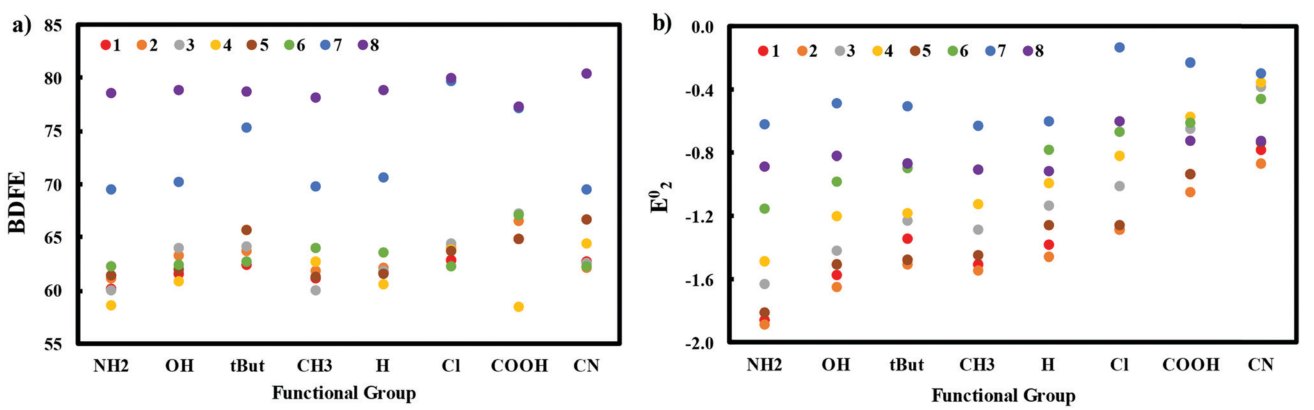


Fig. 5 Effect of EDGs and EWGs on: (a) BDFE, (b) 2nd reduction potential (*vs.* Fc^+/Fc). The chart shows no clear correlation between BDFE and the type of the functional group. On the other hand, the 2nd reduction potential is sensitive to the change of electron density in the ring. This results because after protonation, the positive charge delocalizes over the ring and the extra electron in the π system becomes more localized on the added proton, thus resulting in a more conjugated system of electrons.

the fact that the π system of the resulting radical ($X^•$) is occupied by one extra electron, which makes it non-aromatic and thus lowers its degree of conjugation. On the other hand, BDFEs are shown to cluster in three regions populated by molecules **1–6**, by **7**, and by **8** and their derivatives, which corresponds to variations in resonance stabilization of the radical intermediate ($X^•$).

Our calculations predict that diazaphospholenes exhibit wide ranges of hydricities (29–66 kcal mol^{-1}) and BDFEs (58–81 kcal mol^{-1}). Exploiting this opportunity would expand their space of applications in various chemistries considerably. For instance, CO_2 reduction requires a hydride donor with a hydricity below formate's (43 kcal mol^{-1} in acetonitrile).[52] For this application, molecules **4** and **4-OH** have the least negative reduction potentials while surpassing the required hydricity threshold. Stronger hydrides, such as those based on **2** and **5**, could be implemented to enhance HT kinetics, but at the expense of more negative reduction potentials and thus less efficient regeneration.

Conclusion

In this work, we have explored diazaphospholenes as organic reducing agents by screening 64 candidate molecules, based on 8 main structures and 8 functional groups, and analyzing their hydride/hydrogen atom donation catalytic cycles through an electrochemical mechanism. Phosphorous-based hydrides can be hydrogen atom donors and hydride donors. For instance, diazaphospholenes are strong hydrides as their hydricities can be as low as 29 kcal mol^{-1}. They are also good hydrogen atom donors as their BDFEs are comparable to classical hydrogen atom donors. Despite their exceptional hydridic strength relative to other organic hydrides, they have not been used extensively. One reason is that PBHs are sensitive to oxygen and moisture, although this issue has been addressed by using alkoxydiazaphospholene as a precatalyst, which is not sensitive to air or moisture.[31] Additionally, the tradeoff between their thermodynamic HT ability and their regeneration cost means that stronger reducing agents require more negative potentials to recover. While EDGs can strengthen hydride donors, they have no influence on the hydrogen atom donation ability of these hydrides because H atom donation from these molecules is not driven by aromatization/de-aromatization. Aromatic hydrides are stronger hydrides than their non-aromatic counterparts because their HTs are driven by aromatization. Furthermore, their regeneration requires less negative reduction potentials, which makes them more efficient catalysts. This is mainly attributed to their stable radical intermediates ($X^•$) which correspond to their low BDFE and reorganization energy, which are key factors in shifting the tradeoff relation between hydricity and reduction potential. This concept also explains why phosphorus hydrides can be regenerated with less negative reduction potentials than equivalent C-based hydrides. The analysis in this work provides a starting point towards uncovering chemical and electrochemical catalytic applications of diazaphospholenes, which remain relatively unexplored. For instance, they could be applied towards the electrochemical conversion of CO_2 to methanol, hydrogenation of carbonyl compounds, and hydrogen atom donation reactions. More broadly, understanding the effects of the reorganization energy and BDFE on electrochemical properties could be exploited to enhance the efficiency, and thus regeneration cost, of the catalytic implementation of both existing and newly discovered organic hydrides.

Conflicts of interest

There are no conflicts to declare.

References

1 H. C. Brown and S. Krishnamurthy, Forty years of hydride reductions, *Tetrahedron*, 1979, **35**(5), 567–607.
2 H. C. Brown, S. Kim and S. Krishnamurthy, Selective reductions. 26. Lithium triethylborohydride as an exceptionally powerful and selective reducing agent in organic synthesis. Exploration of the reactions with selected organic compounds containing representative functional groups, *J. Org. Chem.*, 1980, **45**(1), 1–12.
3 G. L. Larson and J. L. Fry, Ionic and Organometallic-Catalyzed Organosilane Reductions, *Org. React.*, 2008, **71**, 1–737.
4 S. Ilic, A. Alherz, C. B. Musgrave and K. D. Glusac, Thermodynamic and kinetic hydricities of metal-free hydrides, *Chem. Soc. Rev.*, 2018, **47**(8), 2809–2836.
5 E. S. Wiedner, M. B. Chambers, C. L. Pitman, R. M. Bullock, A. J. Miller and A. M. Appel, Thermodynamic hydricity of transition metal hydrides, *Chem. Rev.*, 2016, **116**(15), 8655–8692.
6 D. L. DuBois, Development of molecular electrocatalysts for energy storage, *Inorg. Chem.*, 2014, **53**(8), 3935–3960.
7 Y. Kan and Q. Zhang, Transition Metal Complexes for Hydrogen Activation, *Nanostructured Materials for Next-Generation Energy Storage and Conversion*, Springer, 2017, pp. 43–84.
8 R. B. Gordon, M. Bertram and T. E. Graedel, Metal stocks and sustainability, *Proc. Natl. Acad. Sci. U. S. A.*, 2006, **103**(5), 1209–1214.
9 C.-H. Lim, A. M. Holder and C. B. Musgrave, Mechanism of homogeneous reduction of CO_2 by pyridine: proton relay in aqueous solvent and aromatic stabilization, *J. Am. Chem. Soc.*, 2012, **135**(1), 142–154.
10 J. M. Berg, J. L. Tymoczko and L. Stryer, *Biochemistry*, W H Freeman, New York, 5th edn, 2002, ch. 19.
11 X.-Q. Zhu and C.-H. Wang, Hydride affinity scale of various substituted arylcarbeniums in acetonitrile, *J. Phys. Chem. A*, 2010, **114**(50), 13244–13256.
12 J. Cheng, K. L. Handoo and V. D. Parker, Hydride affinities of carbenium ions in acetonitrile and dimethyl sulfoxide solution, *J. Am. Chem. Soc.*, 1993, **115**(7), 2655–2660.
13 C.-H. Lim, S. Ilic, A. Alherz, B. T. Worrell, S. S. Bacon, J. T. Hynes, K. D. Glusac and C. B. Musgrave,

Benzimidazoles as Metal-Free and Recyclable Hydrides for CO_2 Reduction to Formate, *J. Am. Chem. Soc.*, 2018, **141**(1), 272–280.

14 A. Alherz, C.-H. Lim, Y.-C. Kuo, P. Lehman, J. Cha, J. T. Hynes and C. B. Musgrave, Renewable Hydride Donors for the Catalytic Reduction of CO_2: A Thermodynamic and Kinetic Study, *J. Phys. Chem. B*, 2018, **122**(44), 10179–10189.

15 C.-H. Lim, A. M. Holder, J. T. Hynes and C. B. Musgrave, Dihydropteridine/pteridine as a 2H +/2e–redox mediator for the reduction of CO_2 to methanol: A computational study, *J. Phys. Chem. B*, 2017, **121**(16), 4158–4167.

16 C.-H. Lim, A. M. Holder, J. T. Hynes and C. B. Musgrave, Reduction of CO_2 to methanol catalyzed by a biomimetic organo-hydride produced from pyridine, *J. Am. Chem. Soc.*, 2014, **136**(45), 16081–16095.

17 C.-H. Lim, A. M. Holder, J. T. Hynes and C. B. Musgrave, Catalytic reduction of CO_2 by renewable organohydrides, *J. Phys. Chem. Lett.*, 2015, **6**(24), 5078–5092.

18 S. Ilic, P. U. Kadel, Y. Basdogan, J. A. Keith and K. D. Glusac, Thermodynamic hydricities of biomimetic organic hydride donors, *J. Am. Chem. Soc.*, 2018, **140**(13), 4569–4579.

19 X.-M. Zhang, J. W. Bruno and E. Enyinnaya, Hydride affinities of arylcarbenium ions and iminium ions in dimethyl sulfoxide and acetonitrile, *J. Org. Chem.*, 1998, **63**(14), 4671–4678.

20 Z. M. Heiden and A. P. Lathem, Establishing the hydride donor abilities of main group hydrides, *Organometallics*, 2015, **34**(10), 1818–1827.

21 S. Ilic, A. Alherz, C. B. Musgrave and K. D. Glusac, Importance of proton-coupled electron transfer in cathodic regeneration of organic hydrides, *Chem. Commun.*, 2019, **55**(39), 5583–5586.

22 S. Burck, D. Gudat, M. Nieger and W.-W. Du Mont, P-hydrogen-substituted 1, 3, 2-diazaphospholenes: Molecular hydrides, *J. Am. Chem. Soc.*, 2006, **128**(12), 3946–3955.

23 D. Gudat, Diazaphospholenes: N-Heterocyclic phosphines between molecules and Lewis pairs, *Acc. Chem. Res.*, 2010, **43**(10), 1307–1316.

24 D. Gudat, A. Haghverdi and M. Nieger, Umpolung of P–H Bonds, *Angew. Chem., Int. Ed.*, 2000, **39**(17), 3084–3086.

25 C. C. Chong, H. Hirao and R. Kinjo, A Concerted Transfer Hydrogenolysis: 1, 3, 2-Diazaphospholene-Catalyzed Hydrogenation of N·N Bond with Ammonia–Borane, *Angew. Chem., Int. Ed.*, 2014, **53**(13), 3342–3346.

26 B. Rao, C. C. Chong and R. Kinjo, Metal-Free Regio-and Chemoselective hydroboration of pyridines catalyzed by 1, 3, 2-diazaphosphenium triflate, *J. Am. Chem. Soc.*, 2018, **140**(2), 652–656.

27 C. C. Chong and R. Kinjo, Hydrophosphination of CO_2 and Subsequent Formate Transfer in the 1, 3, 2-Diazaphospholene-Catalyzed N-Formylation of Amines, *Angew. Chem., Int. Ed.*, 2015, **54**(41), 12116–12120.

28 J. Zhang, J.-D. Yang and J.-P. Cheng, Diazaphosphinanes as hydride, hydrogen atom, proton or electron donors under transition-metal-free conditions: thermodynamics, kinetics, and synthetic applications, *Chem. Sci.*, 2020, **11**(14), 3672–3679.

29 J. Zhang, J. D. Yang and J. P. Cheng, A Nucleophilicity Scale for the Reactivity of Diazaphospholenium Hydrides: Structural Insights and Synthetic Applications, *Angew. Chem.*, 2019, **131**(18), 6044–6048.

30 J. Zhang, J.-D. Yang and J.-P. Cheng, Recent progress in reactivity study and synthetic application of N-heterocyclic phosphorus hydrides, *Natl. Sci. Rev.*, 2020, **8**(4), nwaa253.

31 M. R. Adams, C. H. Tien, B. S. Huchenski, M. J. Ferguson and A. W. Speed, Diazaphospholene precatalysts for imine and conjugate reductions, *Angew. Chem., Int. Ed.*, 2017, **56**(22), 6268–6271.

32 Y. Zhao and D. G. Truhlar, The M06 suite of density functionals for main group thermochemistry, thermochemical kinetics, noncovalent interactions, excited states, and transition elements: two new functionals and systematic testing of four M06-class functionals and 12 other functionals, *Theor. Chem. Acc.*, 2008, **120**(1–3), 215–241.

33 A. V. Marenich, C. J. Cramer and D. G. Truhlar, Universal solvation model based on solute electron density and on a continuum model of the solvent defined by the bulk dielectric constant and atomic surface tensions, *J. Phys. Chem. B*, 2009, **113**(18), 6378–6396.

34 M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. V. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, J. L. Sonnenberg, D. Williams-Young, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, V. G. Zakrzewski, J. Gao, N. Rega, G. Zheng, W. Liang, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, K. Throssell, J. A. Montgomery Jr., J. E. Peralta, F. Ogliaro, M. J. Bearpark, J. J. Heyd, E. N. Brothers, K. N. Kudin, V. N. Staroverov, T. A. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A. P. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, J. M. Millam, M. Klene, C. Adamo, R. Cammi, J. W. Ochterski, R. L. Martin, K. Morokuma, O. Farkas, J. B. Foresman and D. J. Fox, *Gaussian 16, Revision B.01*, Gaussian, Inc., Wallingford CT, 2016.

35 C. J. Cramer and D. G. Truhlar, Implicit solvation models: equilibria, structure, spectra, and dynamics, *Chem. Rev.*, 1999, **99**(8), 2161–2200.

36 J. Tomasi and M. Persico, Molecular interactions in solution: an overview of methods based on continuous distributions of the solvent, *Chem. Rev.*, 1994, **94**(7), 2027–2094.

37 S. Miertuš, E. Scrocco and J. Tomasi, Electrostatic interaction of a solute with a continuum. A direct utilizaion of AB initio molecular potentials for the prevision of solvent effects, *Chem. Phys.*, 1981, **55**(1), 117–129.

38 M. Cossi, N. Rega, G. Scalmani and V. Barone, Energies, structures, and electronic properties of molecules in solution with the C-PCM solvation model, *J. Comput. Chem.*, 2003, **24**(6), 669–681.

39 J. Shi, X.-Y. Huang, H.-J. Wang and Y. Fu, Hydride Dissociation Energies of Six-Membered Heterocyclic Organic Hydrides Predicted by ONIOM-G4 Method, *J. Chem. Inf. Model.*, 2012, **52**(1), 63–75.

40 X.-J. Qi, Y. Fu, L. Liu and Q.-X. Guo, Ab initio calculations of thermodynamic hydricities of transition-metal hydrides in acetonitrile, *Organometallics*, 2007, **26**(17), 4197–4203.

41 J. T. Muckerman, P. Achord, C. Creutz, D. E. Polyansky and E. Fujita, Calculation of thermodynamic hydricities and the design of hydride donors for CO_2 reduction, *Proc. Natl. Acad. Sci. U. S. A.*, 2012, **109**(39), 15657–15662.

42 G. Gritzner and J. Kuta, Recommendations on reporting electrode potentials in nonaqueous solvents (Recommendations 1983), *Pure Appl. Chem.*, 1984, **56**(4), 461–466.

43 M. Namazian, C. Y. Lin and M. L. Coote, Benchmark calculations of absolute reduction potential of ferricinium/ferrocene couple in nonaqueous solutions, *J. Chem. Theory Comput.*, 2010, **6**(9), 2721–2725.

44 C. P. Kelly, C. J. Cramer and D. G. Truhlar, Single-ion solvation free energies and the normal hydrogen electrode potential in methanol, acetonitrile, and dimethyl sulfoxide, *J. Phys. Chem. B*, 2007, **111**(2), 408–422.

45 V. D. Parker, K. L. Handoo, F. Roness and M. Tilset, Electrode potentials and the thermodynamics of isodesmic reactions, *J. Am. Chem. Soc.*, 1991, **113**(20), 7493–7498.

46 F. Bordwell, J. P. Cheng and J. A. Harrelson, Homolytic bond dissociation energies in solution from equilibrium acidity and electrochemical data, *J. Am. Chem. Soc.*, 1988, **110**(4), 1229–1231.

47 R. B. Weerasooriya, J. L. Gesiorski, A. Alherz, S. Ilic, G. N. Hargenrader, C. B. Musgrave and K. D. Glusac, Kinetics of Hydride Transfer from Catalytic Metal-Free Hydride Donors to CO_2, *J. Phys. Chem. Lett.*, 2021, **12**, 2306–2311.

48 C. Hansch and A. Leo, *Substituent constants for correlation analysis in chemistry and biology*, Wiley, 1979.

49 L. P. Hammett, The Effect of Structure upon the Reactions of Organic Compounds. Benzene Derivatives, *J. Am. Chem. Soc.*, 1937, **59**(1), 96–103.

50 J. Zhang, J.-D. Yang and J.-P. Cheng, Exploiting the radical reactivity of diazaphosphinanes in hydrodehalogenations and cascade cyclizations, *Chem. Sci.*, 2020, **11**(18), 4786–4790.

51 J. Zhang, J.-D. Yang and J.-P. Cheng, Diazaphosphinyl radical-catalyzed deoxygenation of α-carboxy ketones: a new protocol for chemo-selective C–O bond scission via mechanism regulation, *Chem. Sci.*, 2020, **11**(32), 8476–8481.

52 D. L. DuBois and D. E. Berning, Hydricity of transition-metal hydrides and its role in CO_2 reduction, *Appl. Organomet. Chem.*, 2000, **14**(12), 860–862.

